

✱ VISTEXT-MOSQUITO: A Unified Multimodal Dataset for Visual Detection, Segmentation, and Textual Explanation on Mosquito Breeding Sites

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Abstract

*Mosquito-borne diseases pose a major global health risk, requiring early detection and proactive control of breeding sites to prevent outbreaks. In this paper, we present VISTEXT-MOSQUITO, a multimodal dataset that integrates visual and textual data to support automated detection, segmentation, and explanation for mosquito breeding site analysis. The dataset includes 1,828 annotated images for object detection, 142 images for water surface segmentation, and natural language explanation texts linked to each image. The YOLOv9s model achieves the highest precision of 0.92926 and mAP@50 of 0.92891 for object detection, while YOLOv11n-Seg reaches a segmentation precision of 0.91587 and mAP@50 of 0.79795. For textual explanation generation, we tested a range of large vision-language models (LVLMS) in both zero-shot and few-shot settings. Our fine-tuned **Mosquito-LLaMA3-8B** model achieved the best results, with a final loss of 0.0028, a BLEU score of 54.7, BERTScore of 0.91, and ROUGE-L of 0.85. This dataset and model framework emphasize the theme “Prevention is Better than Cure”, showcasing how AI-based detection can proactively address mosquito-borne disease risks. The dataset and implementation code are publicly available at GitHub: <https://github.com/adnanul-islam-jisun/VisText-Mosquito>*

1. Introduction

Mosquito-borne diseases continue to be a leading cause of illness and death worldwide, particularly affecting low- and middle-income countries. According to the World Health Organization (WHO), approximately 700 million people are affected by mosquito-borne illnesses every year, resulting in over one million deaths [19]. Diseases such as Malaria,

Dengue Fever, Zika Virus, and Chikungunya are transmitted by mosquito vectors that breed primarily in stagnant water [20]. The economic burden associated with these diseases is also substantial; for instance, malaria alone is estimated to cost African economies over \$12 billion annually due to healthcare expenses and lost productivity [4].

Despite ongoing efforts to combat these diseases, the incidence of dengue has increased 30-fold in the past 50 years, with nearly 390 million cases reported annually [9]. The rapid pace of urbanization and poor water management in many regions further exacerbate the problem by creating ideal conditions for mosquito breeding [2]. Traditional mosquito control methods, including manual inspection and elimination of breeding sites, are labor-intensive, time-consuming, and often infeasible in large or inaccessible areas [10]. These challenges underscore the urgent need for scalable, technology-driven approaches to detect and manage mosquito habitats more efficiently.

Although computer vision have made significant strides, the task of accurately detecting mosquito breeding places and analyzing water surfaces remains challenging [5]. Existing models often underperform due to a lack of specialized datasets capable of handling both detection and segmentation tasks [21]. Moreover, most datasets are unimodal, lacking contextual explanations that could aid in human interpretation or decision-making [13]. This absence of interpretability limits the utility of AI models in real-world public health scenarios, where both performance and explainability are critical.

To address these gaps, we present VISTEXT-MOSQUITO, the first multimodal benchmark for mosquito breeding site analysis. Our dataset integrates three complementary components: (i) object detection with 1,828 images and 3,752 bounding boxes across five container types (coconut, exocarp, vase, tire, drain, inlet, and bottle); (ii) water surface seg-

mentation with 142 high-resolution images and 253 pixel-level masks; and (iii) natural language explanation with automatically generated and human-validated explanations that capture visual cues linked to mosquito breeding risk. Beyond dataset creation, we perform a comprehensive benchmarking study of state-of-the-art detectors, segmenters, and large vision–language models. Finally, we fine-tune LLaMA3-8B-Vision on our explanation corpus, yielding Mosquito-LLaMA3-8B, which achieves state-of-the-art BLEU, ROUGE-L, and BERTScore, demonstrating the value of domain adaptation for multimodal understanding. Our key contributions are as follows:

- (1) We introduce VISTEXT-MOSQUITO, the first multimodal dataset that unifies object detection, water segmentation, and visual–linguistic explanation for mosquito breeding site analysis.
- (2) We provide high-quality annotations across visual and textual modalities, including bounding boxes, segmentation masks, and textual explanations for explainable AI, which are all human-annotated and validated.
- (3) We benchmark a wide spectrum of detectors, segmenters, and VLMs under zero-shot and few-shot setups, revealing the limitations of unimodal approaches.
- (4) We develop Mosquito-LLaMA3-8B, a domain-adapted VLM that sets a new baseline for multimodal reasoning in public health surveillance.

2. Related Works

Mosquito-borne diseases continue to pose serious public health threats globally, especially in tropical and subtropical regions [24]. Effective surveillance of mosquito breeding habitats is essential for early intervention and disease prevention. In recent years, advances in computer vision and machine learning have enabled automated detection of mosquito habitats using images and spatial data.

Several datasets have been proposed to support AI-based mosquito breeding site detection. The MosquitoFusion dataset [21] offers annotated real-world images capturing mosquitoes, larvae, swarms, and breeding sites across diverse contexts. Mehra et al. [16] created an entomological image dataset for training lightweight object detection models under challenging lighting and occlusion conditions. Similarly, Chathura et al. [23] presented a drone-captured image dataset that includes standing water sources and vegetation typically associated with mosquito breeding, enhancing the generalizability of deep learning models to outdoor environments. Object detection architectures such as YOLOv5, YOLOv8, and EfficientDet have shown promising results in localizing potential breeding containers, with

YOLOv8 achieving a mean average precision (mAP@50) of 57.1% on the MosquitoFusion dataset [21]. For semantic and instance segmentation, CNN-based models like U-Net and DeepLabV3+ have been used to segment water surfaces from drone and satellite imagery with pixel-level accuracy [16, 18]. Recent transformer-based models like DETR [17] and the Segment Anything Model (SAM) [12] offer superior segmentation quality by learning rich visual features from complex environments.

Geospatial AI techniques have been applied to UAV and satellite imagery for spatial mapping of mosquito breeding hotspots. For instance, GIS-based models using remote sensing data and LIDAR have enabled high-resolution risk mapping of urban habitats [5, 22]. Studies have also explored topographical features, vegetation indices, and hydrological data to predict stagnant water accumulation areas likely to host larvae [11]. Moreover, ensemble machine learning methods (e.g., Random Forest, XGBoost) using environmental and epidemiological data have improved mosquito vector prediction models, especially when coupled with weather and public health surveillance data [3]. In the multimodal domain, progress has been slower. Most existing works focus solely on image-based detection or segmentation, with limited incorporation of textual or reasoning-based outputs. However, natural language generation (NLG) in vision-language tasks has gained popularity, especially with models like BLIP [14], LLaVA [15], and Flamingo [1], which generate human-like captions or justifications. These models have been applied to medical imaging, environmental monitoring, and accessibility applications, showing promise for tasks requiring explainability and human alignment.

Despite these advancements, notable limitations persist. Many datasets are restricted to lab environments or synthetic scenes, lacking the visual complexity of urban and peri-urban breeding sites. Furthermore, most existing models treat breeding site identification as a visual-only task, ignoring the value of language-based explanations for public health decision-making and education. To the best of our knowledge, no prior work has introduced a multimodal dataset that jointly supports object detection, water surface segmentation, and visual explanation tailored to mosquito breeding site analysis. This represents a significant research gap, as integrating vision and language could boost both interpretability and field applicability of AI models for mosquito vector control.

3. Dataset Construction Process

3.1. Data Collection

The data collection process is designed to ensure diversity, accuracy, and real-world relevance in capturing mosquito breeding sites and water surfaces. High-quality images are

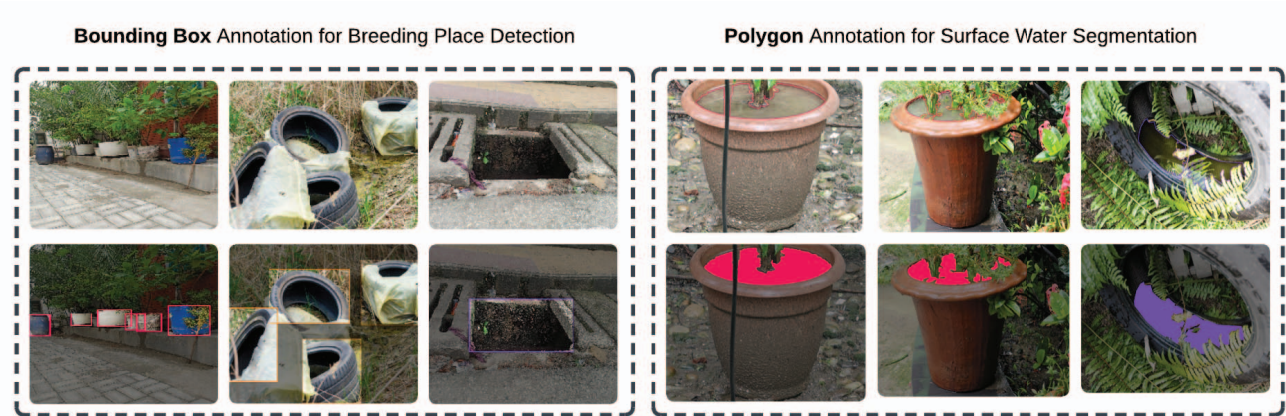


Figure 1. Fully tagged and labeled images for breeding site detection and surface water segmentation

collected from various regions across Bangladesh, covering diverse breeding habitats under both daylight (8 AM–5 PM) and nighttime conditions to enhance dataset variability. To improve model generalization, multiple images are taken from different angles and distances (1–3 meters), ensuring a detailed visual representation. Both natural and artificial breeding sites are documented, though challenges such as unpredictable weather and difficult terrain occasionally impacted data collection. Ethical considerations are prioritized by obtaining permission from local authorities and property owners. The process remains non-invasive, avoiding harm to natural habitats or disruptions to local communities. Anonymization techniques are applied to protect sensitive location details.

The initial dataset comprises 1,828 images with 3,752 annotations for breeding place detection and 142 images with 253 annotations for water surface segmentation. This ensures a diverse, comprehensive foundation for training models on mosquito habitat detection and surface segmentation. Additionally, a text modality is included in the dataset to enable multimodal analysis. This dataset contains 3,762 instances, each associated with an image and annotated with three text fields: **(a) Question:** A binary question asking whether the image shows a mosquito breeding site. **(b) Response:** A 'Yes' or 'No' answer (3,748 "Yes" and 14 "No" responses). **(c) Textual Explanation:** A short free-text explanation justifying the response. The average length of the textual explanation statements is approximately 36 tokens. This textual annotation provides semantic context and interpretability, significantly enhancing the dataset's capacity for explainable AI.

3.2. Data Preprocessing

The data preprocessing involves annotation, transformations, and augmentation to enhance the dataset. All images are manually annotated using the Roboflow [7] plat-

form, ensuring precise labeling of mosquito breeding sites and water surfaces. Figure 1 shows examples of the annotated images. The following preprocessing steps are applied to each image: **(a) Auto-Orient:** Images are auto-oriented to correct any device orientation inconsistencies. **(b) Resize:** All images are resized to 640x640 pixels for uniform input shape. **(c) Auto-Adjust Contrast:** The contrast is automatically adjusted to enhance visual clarity. To improve model robustness, augmentation techniques are applied: **(a) Flip:** Horizontal flips doubled the dataset size by varying object orientations. **(b) Rotation:** Random rotations introduce alignment variations. **(c) Brightness Adjustment:** Image brightness is varied to simulate real-world lighting conditions. As a result of these augmentations, the total number of images in the dataset increases to 4,425 for the detection part and 331 for the segmentation part. This augmentation strategy significantly enhances the dataset's variability, ensuring that the models trained on this dataset would be more robust and capable of generalizing well to unseen data. For the text modality, the binary responses ("Yes" or "No") and the accompanying explanation statements were initially generated using the GPT-4o model. To ensure high-quality annotations, all generated responses are subsequently curated and validated by human annotators. This semi-automated annotation workflow allows efficient dataset expansion while preserving semantic integrity and contextual accuracy.

3.3. Distribution Analysis and Folder Structure

The breeding place detection subset of the dataset comprises a total of 1,828 images with 3,752 annotations distributed across five classes. The selected object categories were derived from existing literature on mosquito ecology and vector-control studies, as well as environmental field observations identifying common stagnant-water containers in endemic regions [6, 8]. The class-wise distribu-

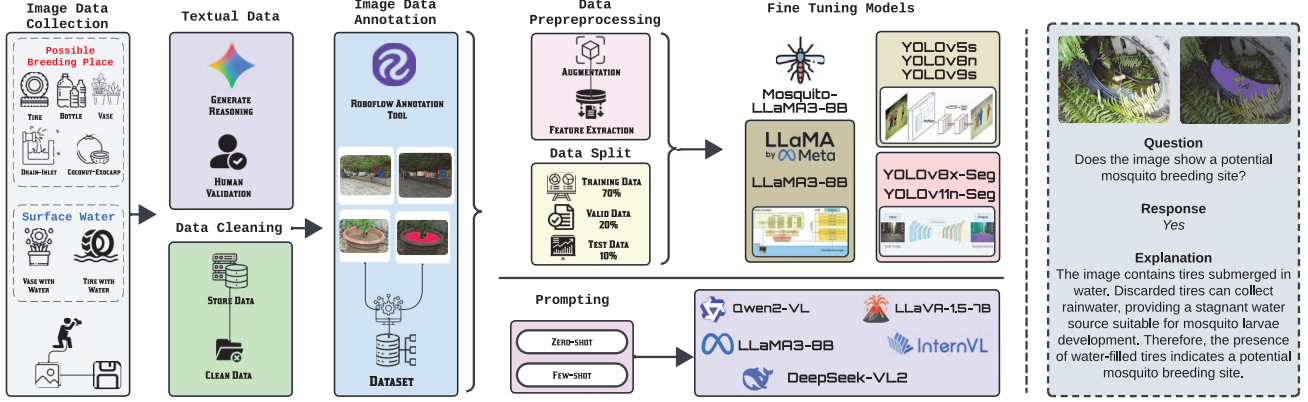


Figure 3. Overview of the VISTEXT-MOSQUITO pipeline, which involves several key steps: data collection, annotation, preprocessing, and model training. It supports object detection, water surface segmentation, and explanation generation for mosquito breeding site analysis.

our specific domain. To support multimodal understanding, we explore a wide range of large vision-language models for the task of generating natural language explanations based on images. These models are tasked with answering a yes/no question and generating free-form textual justifications grounded in visual evidence. We evaluate both open-source and closed-source VLMs. From the closed-source domain, we test Gemini-2.5-Flash, which is a proprietary model. For open-source alternatives, we evaluate LLaVA-1.5-7B, LLaMA3-8B-Vision, DeepSeek-VL2, InternVL-4B, and Qwen2-VL.

4.3. Prompting Strategies

Each vision-language model is tested under two prompting strategies: zero-shot and few-shot. In the zero-shot setting, the model is given only the input image and question, without any in-context examples. This setup evaluates the model’s out-of-the-box generalization capacity to unseen prompts. In the few-shot setting, we include three image-question-explanation pairs as exemplars in the prompt to guide the model.

4.4. Supervised Fine-Tuning

Beyond prompting, we perform supervised fine-tuning using the LLaMA3-8B-Vision model on the VISTEXT-MOSQUITO dataset, which is named Mosquito-LLaMA3-8B. During fine-tuning, the model learns to generate explanation text conditioned on the image content and question prompt using a cross-entropy loss over token probabilities. The model is optimized using the AdamW optimizer with a learning rate of 5×10^{-5} and trained using a batch size of 16. Training proceeds for 3 epochs. This process enables the model to capture domain-specific visual-textual associations and produce reliable and interpretable textual explanations.

4.5. Evaluation Metrics

To measure the effectiveness of our models, we adopt both vision-based and language-based evaluation metrics. For object detection and segmentation tasks, we report precision, recall, and mean Average Precision at Intersection over Union (IoU) threshold 0.5 (mAP@50). These metrics provide insight into how accurately the models detect and localize breeding sites and water surfaces. For the textual explanation generation task, we use a combination of BLEU, ROUGE-L, and BERTScore to evaluate the quality of generated text. BLEU captures n-gram overlap with reference textual explanation, ROUGE-L assesses longest common subsequences to reflect fluency and relevance, while BERTScore leverages contextual embeddings to capture semantic similarity.

5. Result Analysis

5.1. Object Detection Performance

To detect potential mosquito breeding sites, we fine-tune and evaluate four object detection models, YOLOv5s, YOLOv8n, YOLOv9s, and RT-DETR ResNet-101, on 1,828 annotated images across five object classes. The detailed performance is summarized in Table 2.

The YOLOv9s model achieves the highest precision of 0.92926 and mAP@50 of 0.92891, demonstrating its superior ability to accurately localize and classify breeding-related objects. In contrast, YOLOv5s offers the most balanced performance, with a strong precision of 0.91514 and the highest recall of 0.87595. This makes it particularly well-suited for applications where minimizing false negatives is essential, such as early-stage mosquito surveillance. The RT-DETR ResNet-101 model achieves a precision of 0.88935, recall of 0.82778, and mAP@50 of 0.87871, showing competitive detection capability but lower overall

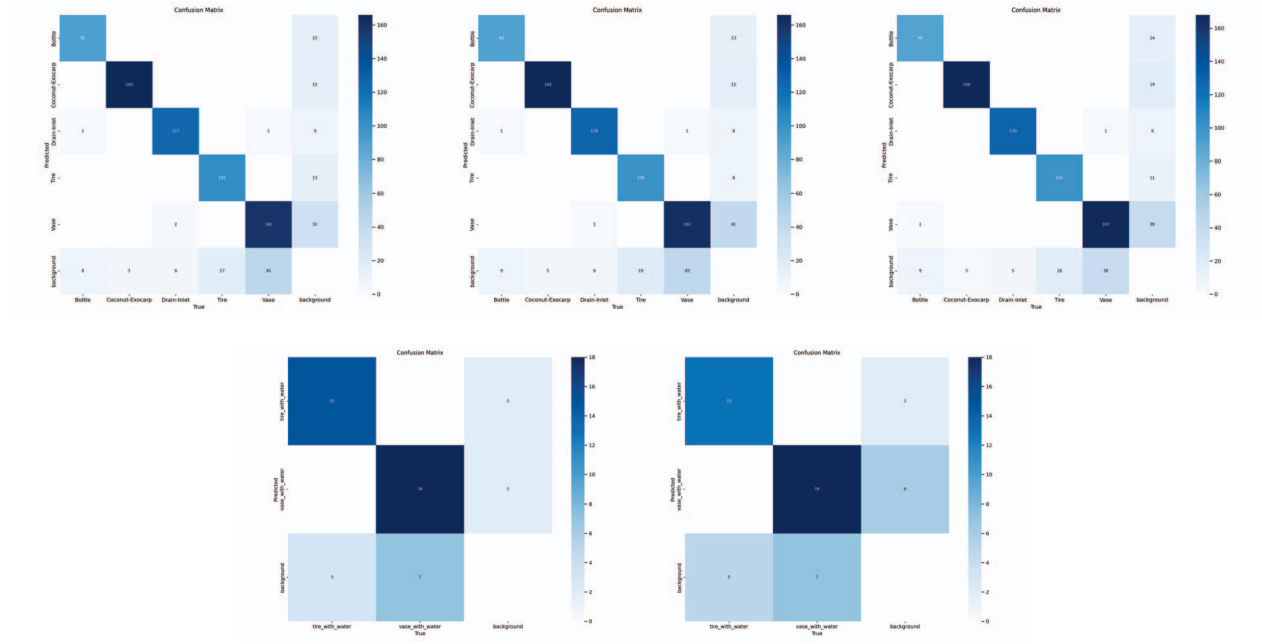


Figure 4. Confusion matrices for (Top) object detection and (Bottom) segmentation tasks.

Model	Precision	Recall	mAP@50
YOLOv5s	0.91514	0.87595	0.92400
YOLOv8n	0.89028	0.87314	0.90817
YOLOv9s	0.92926	0.86011	0.92891
RT-DETR ResNet-101	0.88935	0.82778	0.87871

Table 2. Object Detection Model Performance

performance than the YOLO-based variants on this dataset. Meanwhile, YOLOv8n shows slightly stronger recall and mAP@50 than RT-DETR, with a precision of 0.89028, recall of 0.87314, and mAP@50 of 0.90817. These results suggest that the architectural improvements in YOLOv9s contribute to meaningful gains in real-world breeding site detection tasks.

5.2. Water Surface Segmentation Performance

The segmentation task is focused on identifying water surfaces in objects like vases and tires, critical indicators of mosquito breeding potential. We fine-tuned three advanced models, YOLOv8x-Seg, YOLOv11n-Seg, and Mask R-CNN (ResNet-101 FPN), on 142 images annotated for `vase_with_water` and `tire_with_water`. Table 3 presents their performance.

YOLOv11n-Seg consistently achieves the strongest overall performance across all three evaluation metrics. Specifically, it attains the highest precision of 0.91587, compared to 0.89372 for YOLOv8x-Seg and 0.90142 for Mask R-CNN (ResNet-101 FPN). It also records the

Model	Precision	Recall	mAP@50
YOLOv8x-Seg	0.89372	0.73074	0.79345
YOLOv11n-Seg	0.91587	0.77201	0.79795
Mask R-CNN	0.90142	0.74863	0.79511

Table 3. Segmentation Model Performance

best recall of 0.77201, indicating a stronger ability to capture true positive water-surface regions without missing relevant segments. In terms of mAP@50, YOLOv11n-Seg again performs best with 0.79795, while Mask R-CNN achieves a competitive score of 0.79511, slightly surpassing YOLOv8x-Seg.

The inclusion of Mask R-CNN provides a strong two-stage segmentation baseline for comparison with the YOLO-based one-stage segmentation models. Its competitive precision and mAP@50 suggest that region-based segmentation remains effective for capturing structured water-holding regions such as vase interiors and tire cavities. However, YOLOv11n-Seg demonstrates the best balance between localization quality and segmentation consistency, making it the most robust model for pixel-level mosquito habitat identification in visually challenging real-world environments with clutter, shadows, and partial occlusion.

5.3. Multimodal Explanation Performance

For the textual explanation task, we fine-tune the LLaMA3-8B-Vision model using image-question-explanation triplets curated from the VISTEXT-MOSQUITO

Model	Zero-shot			Few-shot		
	BLEU	BERTScore	ROUGE-L	BLEU	BERTScore	ROUGE-L
Open-source						
LLaVA-1.5-7B	38.2	0.66	0.65	39.1	0.68	0.69
LLaMA3-8B-Vision	40.2	0.71	0.68	42.8	0.78	0.72
DeepSeek-VL2	36.7	0.64	0.63	38.2	0.73	0.66
InternVL-4B	33.5	0.52	0.57	35.4	0.56	0.60
Qwen2-VL	31.6	0.55	0.60	29.3	0.50	0.55
Closed-source						
Gemini-2.5-Flash	45.9	0.78	0.73	47.5	0.83	0.77
GPT-4o mini	43.1	0.80	0.75	46.2	0.85	0.79
Fine-tuned						
	BLEU	BERTScore	ROUGE-L			
Mosquito-LLaMA3-8B (Ours)	54.7	0.91	0.85			

Table 4. Zero-shot, Few-shot, and Fine-tuned Textual Explanation Performance Across Vision–Language Models. The fine-tuned model is evaluated only after task-specific adaptation.

dataset. The model is trained to generate coherent and semantically grounded justifications that explain whether an image depicts a mosquito breeding site and why. During training, it learns to associate domain-specific visual patterns, such as standing water in tires, vases, or drains, with meaningful language that reflects expert-level explanation. After three epochs, the model converges with a final training loss of 0.0028, suggesting stable training and strong multimodal alignment. To evaluate the generated justifications, we adopt standard natural language generation metrics including BLEU, BERTScore, and ROUGE-L. Our fine-tuned model, referred to as *Mosquito-LLaMA3-8B*, achieves a BLEU score of 54.7, a BERTScore of 0.91, and a ROUGE-L of 0.85. These results indicate strong n-gram precision, semantic similarity, and structural alignment with human-written reference texts. Qualitative inspection further supports these findings: the model effectively integrates visual context, such as the presence of water, container types, and environmental cues, into fluent, human-readable rationales. Beyond supervised fine-tuning, we assess the generalization and adaptation abilities of other vision-language models under both zero-shot and few-shot prompting setups. In the zero-shot setting, models receive only an image and a prompt without any additional context. In the few-shot setting, each model is shown three image-question-explanation examples before generating responses for unseen inputs. This experimental design enables us to evaluate both generalization from pretraining and responsiveness to in-context learning. Table 4 summarizes the results. Overall, few-shot prompting improved performance across most models, with notable increases in BLEU, BERTScore, and ROUGE-L. For instance, *LLaMA3-8B-Vision* improved from 40.2 to 42.8 in BLEU, and from 0.71 to 0.78 in BERTScore.

LLaVA-1.5-7B and *DeepSeek-VL2* showed modest but consistent improvements across all metrics, while even the smaller *InternVL-4B* gained from few-shot examples, despite its relatively lower baseline. Interestingly, the closed-source *Gemini-2.5-Flash* displayed only minor improvements, possibly due to its already strong zero-shot capabilities limiting headroom for further gain. A notable exception to this trend is *Qwen2-VL*, which experienced a performance decline in all three metrics under few-shot prompting, dropping from 31.6 to 29.3 in BLEU, from 0.55 to 0.50 in BERTScore, and from 0.60 to 0.55 in ROUGE-L. This suggests that *Qwen2-VL* may be particularly sensitive to prompt construction or lacks sufficient in-context learning capabilities for domain-specific textual explanation tasks. In contrast, our *Mosquito-LLaMA3-8B* model exhibits significant improvements across all metrics. Compared to its own zero-shot variant, it achieves a +13.3 point gain in BLEU, +0.04 in BERTScore, and +0.07 in ROUGE-L. These results highlight the clear advantage of supervised domain-specific fine-tuning over reliance on general pretraining or prompt-based learning. Overall, our findings indicate that while few-shot prompting enhances textual explanation capabilities in most models, fine-tuning remains essential for achieving high-accuracy, context-sensitive reasoning in public health surveillance applications.

5.4. Ablation Study

To understand the effect of different training configurations on textual explanation performance, we conduct an ablation study on the fine-tuned *Mosquito-LLaMA3-8B* model. We vary critical training hyperparameters, including the number of training epochs, learning rate, and the number of image-question-explanation triplets used during training.

Configuration	BLEU	BERTScore	ROUGE-L	Final Loss
Baseline (3 epochs, full data)	54.7	0.91	0.85	0.0028
1 epoch (underfitting)	46.2	0.82	0.76	0.0085
5 epochs (overfitting)	49.7	0.87	0.81	0.0016
Learning rate 1×10^{-5}	47.5	0.85	0.79	0.0043
Learning rate 1×10^{-4}	48.8	0.86	0.80	0.0039
50% training examples	42.1	0.76	0.72	0.0071
75% training examples	48.3	0.84	0.77	0.0052
No image augmentation	46.8	0.81	0.75	0.0060

Table 5. Ablation Study on Fine-Tuned Mosquito-LLaMA3-8B

The goal is to isolate how each factor contributes to the model’s ability to align visual cues with textual explanations. Table 5 summarizes the results. The baseline model (our final version) is trained for 3 epochs with a learning rate of 5×10^{-5} and the full dataset. Reducing the number of epochs to 1 or increasing it to 5 both result in slightly degraded performance, likely due to underfitting and overfitting, respectively. Similarly, lowering the learning rate delays convergence and results in weaker semantic alignment. Using only 50% of the training examples notably degrades all metrics, demonstrating the importance of dataset size for effective domain-specific textual explanation. The ablation confirms that the final configuration strikes a strong balance between generalization and alignment. Notably, the number of training examples and moderate epoch count are the most influential factors.

6. Discussion

Despite the strong performance of both detection and explanation models on the VISTEXT-MOSQUITO dataset, several challenges and limitations remain, particularly in the context of multimodal understanding. Through qualitative analysis, we observed recurring failure cases among VLMs, especially in zero-shot scenarios. For instance, models like Qwen2-VL and InternVL-4B often failed to generate grounded explanations when visual cues were subtle or occluded. A common example includes misinterpreting a dry tire partially filled with leaves as a mosquito breeding site, where the model incorrectly inferred the presence of water based on contextual noise rather than actual visual evidence. Another frequent failure pattern emerged when visually similar objects led to semantic confusion. VLMs sometimes misclassified objects such as a decorative vase or a plastic bucket as breeding containers even when they lacked water. This suggests that models are often over-reliant on coarse object shape and context rather than finer visual-textual alignment.

Additionally, models struggled when backgrounds were cluttered or when key cues like water surfaces were not prominently visible. These cases highlight the limitations

of current VLMs in precisely localizing fine-grained visual patterns critical to public health applications. The disparity between zero-shot and few-shot performance across models further reinforces the importance of context-aware prompting. Closed-source models like Gemini-2.5-Flash performed reasonably well even without training, but open-source alternatives significantly improved under few-shot settings, indicating their reliance on task-specific cues. However, even in few-shot setups, hallucinations were observed, and some models generated plausible yet factually incorrect textual explanations, such as inferring mosquito larvae without any visible evidence. These findings emphasize the need for further advancements in VLM robustness, particularly in environments with ambiguous visual signals. Incorporating spatial grounding, multimodal attention refinement, and post-hoc factuality checks could improve reliability. Moreover, continued dataset expansion with hard negative samples and edge-case explanation examples may help fine-tune these models for real-world vector surveillance scenarios.

7. Conclusion

The VISTEXT-MOSQUITO dataset establishes a comprehensive benchmark for automated detection, segmentation, and textual explanation on mosquito breeding sites. Through rigorous experiments, we show that YOLOv9s yields the best detection accuracy, while YOLOv11n-Seg excels in water surface segmentation. Our fine-tuned Mosquito-LLaMA3-8B model shows strong performance in generating coherent justifications, bridging visual cues with textual explanation. These results underscore the potential of integrating vision-language models into public health monitoring frameworks. By combining high-performance object recognition with textual explanation, our dataset paves the way for intelligent, interpretable, and scalable tools to support vector control strategies. Future work will expand the dataset across ecological regions, enhance textual explanation diversity, and explore prompt-adaptive models for localized intervention planning.

References

- [1] Jean-Baptiste Alayrac, Jeff Donahue, Paul Luc, Antoine Miech, Ian Barr, et al. Flamingo: a visual language model for few-shot learning. *arXiv preprint arXiv:2204.14198*, 2022. [2](#)
- [2] Iswar Baitharu, Sabita Shroff, Prajna Paramita Naik, and Jayanta Kumar Sahu. Environmental management and sustainable control of mosquito vector: challenges and opportunities. *Molecular identification of mosquito vectors and their management*, pages 129–147, 2021. [1](#)
- [3] Sean P Beeman. *Synthesis of a Multimodal Ecological Model for Scalable, High-Resolution Arboviral Risk Prediction in Florida*. PhD thesis, University of South Florida, 2021. [2](#)
- [4] Leo Braack, A Paulo Gouveia de Almeida, Anthony J Cornel, Robert Swanepoel, and Christiaan De Jager. Mosquito-borne arboviruses of african origin: review of key viruses and vectors. *Parasites & vectors*, 11:1–26, 2018. [1](#)
- [5] Daniel Trevisan Bravo, Gustavo Araujo Lima, Wonder Alexandre Luz Alves, Vitor Pessoa Colombo, Luc Djogbenou, Sergio Vicente Denser Pamboukian, Cristiano Capellani Quaresma, and Sidnei Alves de Araujo. Automatic detection of potential mosquito breeding sites from aerial images acquired by unmanned aerial vehicles. *Computers, Environment and Urban Systems*, 90:101692, 2021. [1](#), [2](#)
- [6] Tadesse Dejenie, Mekonnen Yohannes, and Tsehaye Assmelash. Characterization of mosquito breeding sites in and in the vicinity of tigray microdams. *Ethiopian journal of health sciences*, 21(1):57–66, 2011. [3](#)
- [7] B Dwyer, J Nelson, J Solawetz, et al. Roboflow (version 1.0)[software], 2022. [3](#)
- [8] Ulrike Fillinger, Heleen Sombroek, Silas Majambere, Emiel van Loon, Willem Takken, and Steven W Lindsay. Identifying the most productive breeding sites for malaria mosquitoes in the gambia. *Malaria journal*, 8(1):62, 2009. [3](#)
- [9] Si-Lei Fong, Kum-Thong Wong, and Chong-Tin Tan. Dengue virus infection and neurological manifestations: an update. *Brain*, 147(3):830–838, 2024. [1](#)
- [10] Mooyoung Han, Shervin Hashemi, Sung Hee Joo, and Tschungil Kim. Novel integrated systems for controlling and prevention of mosquito-borne diseases caused by poor sanitation and improper water management. *Journal of Environmental Chemical Engineering*, 4(4):3718–3723, 2016. [1](#)
- [11] Md. Adnanul Islam, Md. Faiyaz Abdullah Sayeedi, Jannatul Ferdous Deepti, Shahanur Rahman Bappy, Safrin Sanzida Islam, and Fahim Hafiz. Mosquitominer: A light weight rover for detecting and eliminating mosquito breeding sites. In *2024 IEEE Region 10 Symposium (TENSYP)*, pages 1–6, 2024. [2](#)
- [12] Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Laura Rolland, Kaiming Gustafson, et al. Segment anything. *arXiv preprint arXiv:2304.02643*, 2023. [2](#)
- [13] Camila Laranjeira, Daniel Andrade, and Jefersson A. dos Santos. Yolov7 for mosquito breeding grounds detection and tracking. In *2023 IEEE International Conference on Image Processing Challenges and Workshops (ICPCW)*, pages 1–5, 2023. [1](#)
- [14] Junnan Li, Dongxu Li, Caiming Xiong, and Steven C. Hoi. Blip: Bootstrapping language-image pre-training for unified vision-language understanding and generation. In *International Conference on Machine Learning (ICML)*, 2022. [2](#)
- [15] Haotian Liu, Chunyuan Li, Yuheng Zhang, and Jianfeng Wang. Visual instruction tuning. *arXiv preprint arXiv:2304.08485*, 2023. [2](#)
- [16] Maanit Mehra, Aditya Bagri, Xiaofan Jiang, and Jorge Ortiz. Image analysis for identifying mosquito breeding grounds. In *2016 IEEE International conference on sensing, communication and networking (SECON workshops)*, pages 1–6. IEEE, 2016. [2](#)
- [17] Pravina Mylvaganam and Maheshi B Dissanayake. Detection of mosquito breeding areas using semantic segmentation. In *Proceedings of the 3rd International Women in Engineering Symposium 2022*, pages 34–36, 2022. [2](#)
- [18] Wesley L Passos, Gabriel M Araujo, Amaro A de Lima, Sergio L Netto, and Eduardo AB da Silva. Automatic detection of aedes aegypti breeding grounds based on deep networks with spatio-temporal consistency. *Computers, Environment and Urban Systems*, 93:101754, 2022. [2](#)
- [19] Adnan I. Qureshi. Chapter 2 - mosquito-borne diseases. In *Zika Virus Disease*, pages 27–45. Academic Press, 2018. [1](#)
- [20] SNR Saleeza, Y Norma-Rashid, and Mohd Sofian-Azirun. Mosquito species and outdoor breeding places in residential areas in malaysia. *Southeast Asian Journal of Tropical Medicine and Public Health*, 44(6):963–969, 2013. [1](#)
- [21] Md. Faiyaz Abdullah Sayeedi, Fahim Hafiz, and Md Ashiqur Rahman. Mosquitofusion: A multiclass dataset for real-time detection of mosquitoes, swarms, and breeding sites using deep learning. In *The Second Tiny Papers Track at ICLR 2024*, 2024. [1](#), [2](#)
- [22] Jared Schenkel, Paul Taele, Daniel Goldberg, Jennifer Horney, and Tracy Hammond. Identifying potential mosquito breeding grounds: Assessing the efficiency of uav technology in public health. *Robotics*, 9(4):91, 2020. [2](#)
- [23] Chathura Suduwella, Akarshani Amarasinghe, Lasith Niroshan, Charith Elvitigala, Kasun De Zoysa, and Chamath Keppetiyagama. Identifying mosquito breeding sites via drone images. In *proceedings of the 3rd workshop on micro aerial vehicle networks, systems, and applications*, pages 27–30, 2017. [2](#)
- [24] World Health Organization. Mosquito-borne diseases, 2022. Accessed: 2025-09-13. [2](#)